EL3 Worked Examples: Introduction to Logarithms

Recording companion for the EL3 CheckIt support video — Math 172, Santa Ana College

How to use this document

- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

What power do I raise the base to in order to get the number inside the logarithm? That power is the value of the logarithm. An exponential equation and a logarithmic equation say the same thing in two different ways.

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Example 1: Convert 1

$$3^4=81$$

Question

Which equation is the logarithmic form of this exponential equation?

- **A.** $\log_3 4=81$
- **B.** $\log_4 81=3$
- **C.** $\log_3 81=4$
- **D.** $\log_{81} 3=4$

Answer and reasoning

The base of the exponential, 3, becomes the base of the log. The exponent, 4, becomes the value of the log, and 81 goes inside. Read it back as a check: $\log_3 81=4$ says "3 to the 4th power is 81." **The logarithmic form is $\log_3 81=4$.**

Example 2: Convert 2

$$7^n = p$$

Question

Which equation is the logarithmic form of this exponential equation?

- **A.** $\log_7 n = p$
- **B.** $\log_p 7 = n$
- **C.** $\log_7 p = n$
- **D.** $\log_n 7 = p$

Answer and reasoning

Letters follow the same pattern as numbers. The base, 7, stays the base. The exponent, n, is the value of the log. The result, p, goes inside the log. Read it back: $\log_7 p = n$ says "7 to the n is p." **The logarithmic form is $\log_7 p = n$.**

Example 3: Convert 3

$$\log_5 125=3$$

Question

Which equation is the exponential form of this logarithmic equation?

- **A.** $3^5=125$
- **B.** $5^{125}=3$
- **C.** $5^3=125$
- **D.** $125^3=5$

Answer and reasoning

The base of the log, 5, is the base of the power. The value of the log, 3, is the exponent. The number inside, 125, is the result. Check it: $5\cdot 5\cdot 5=125$. **The exponential form is $5^3=125$.**

Example 4: Convert 4

$$\log_b a=c$$

Question

Which equation is the exponential form of this logarithmic equation?

- **A.** $a^c=b$
- **B.** $b^c=a$
- **C.** $c^b=a$
- **D.** $b^a=c$

Answer and reasoning

Start at the base, b. The value of the log, c, is the exponent. The number inside, a, is the result. Read it back: $b^c=a$ says "b to the c is a." **The exponential form is $b^c=a$.**

Example 5: Evaluate 1

$$\log_2 32$$

Question

What is the value of this logarithm?

A. 16
B. 64
C. 5
D. 2

Answer and reasoning

Ask: 2 to what power gives 32? Count the factors: $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 32$, so $2^5 = 32$. The value of the logarithm is that exponent. **$\log_2 32 = 5$.**

Example 6: Evaluate 2

$$\log_3(1/9)$$

Question

What is the value of this logarithm?

A. 2
B. −2
C. −9
D. *1/2*

Answer and reasoning

Ask: 3 to what power gives *1/9*? Since $3^2=9$, a negative exponent flips it to $3^{-2}=1/9$. A logarithm can be negative when the number inside is between 0 and 1. **$\log_3(1/9)=-2$.**

Example 7: Evaluate 3

$$\log_{16} 4$$

Question

What is the value of this logarithm?

- **A.** 4
- **B.** 2
- **C.** *1/2*
- **D.** *1/4*

Answer and reasoning

Ask: 16 to what power gives 4? A square root undoes a square, so $16^{1/2}=\sqrt{16}=4$, and the power is *1/2*. Do not flip the base and the number: $\log_4 16$ would be 2, but here the base is 16. **$\log_{16} 4 = 1/2$.**

Example 8: Common Log

$$\log 0.01$$

Question

What is the value of this logarithm?

A. 2
B. −2
C. −100
D. 0.1

Answer and reasoning

With no base written, the base is 10. Ask: 10 to what power gives 0.01? Since $0.01=1/100=1/(10^2)=10^{-2}$, the power is -2. **$\log 0.01=-2$.**

Example 9: Natural Log 1

$$\ln 7 = x$$

Question

Which equation is the exponential form of this logarithmic equation?

- **A.** $e^{7}=x$
- **B.** $7^{x}=e$
- **C.** $e^{x}=7$
- **D.** $x^{e}=7$

Answer and reasoning

ln is a logarithm with base e, so $\ln 7 = x$ means $\log_e 7 = x$. The base is e, the exponent is x, and the result is 7. **The exponential form is $e^{x}=7$.**

Example 10: Natural Log 2

$$\ln(e^5)$$

Question

What is the value of this logarithm?

A. e
B. e^5
C. 5
D. $1/5$

Answer and reasoning

Ask: e to what power gives e^5? The power is 5. The natural log undoes a power of e.
$\ln(e^5)=5$.

Example 11: Slip 1

$$\log 1000 = 3$$

Question

A student says this is true, because no base written means base 10, and 10 cubed is 1000. What is the mistake, if any?

A. With no base written, the base is e, not 10.
B. The answer should be 1000 divided by 10, which is 100.
C. There is no mistake.
D. A log with no base cannot be evaluated at all.

Answer and reasoning

log x with no base written means base 10. Ask: 10 to what power gives 1000? $10^3 = 1000$, so the answer is 3. The other common log, ln, has base e. **There is no mistake.**

Example 12: Slip 2

$$2^5=32$$

$$\log_5 32=2$$

Question

A student converted this equation to logarithmic form. What is the mistake, if any?

A. There is no mistake.
B. The 32 belongs outside the log: $\log_5 2=32$.
C. Logarithms only use base 10, so it is log $32=5$.
D. The base stays the base: $\log_2 32=5$.

Answer and reasoning

In $2^5=32$ the base is 2 and the exponent is 5. A logarithm keeps the base and gives back the exponent: $\log_2 32=5$. The student swapped the base and the exponent. **The logarithmic form is $\log_2 32=5$.**

Example 13: Slip 3

$$\log_4 64=3$$
$$3^4=64$$

Question

A student converted this equation to exponential form. What is the mistake, if any?

A. Raise the base to the value: $4^3=64$.
B. There is no mistake.
C. The 64 belongs in the exponent instead: $4^{64}=3$.
D. Put the 64 in the base instead: $64^3=4$.

Answer and reasoning

The base of the log becomes the base of the power, and the value of the log becomes the exponent: $4^3=64$. Check: $4\cdot 4\cdot 4=64$, but $3^4=81$. **The exponential form is $4^3=64$.**

Example 14: Slip 4

$$\log_3 81 = 27$$

Question

A student divided 81 by 3 and wrote this. What is the mistake, if any?

A. There is no mistake.
B. The answer should be 81 times 3, which is 243.
C. 3 to the 4th is 81, so the log is 4.
D. The answer should be 3, the base.

Answer and reasoning

A logarithm is an exponent. Ask: 3 to what power gives 81? $3^4 = 81$, so the answer is 4. Dividing has nothing to do with it. **The correct value is $\log_3 81 = 4$.**

Example 15: Slip 5

$$\log_2(1/8)=3$$

Question

A student wrote this. What is the mistake, if any?

A. The answer should be one third, the flip of 3.
B. There is no mistake.
C. The log of a fraction is undefined, so there is no answer.
D. One eighth is 2 to the negative 3, so it is negative 3.

Answer and reasoning

Write the number inside as a power of 2: $1/8=1/(2^3)=2^{-3}$. So the exponent is -3. Since $2^3=8$, not $1/8$, the student's answer is the log of 8. **The correct value is -3.**

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